

DH²A-KT: Leakage-Audited Knowledge Construction, Time-Aware Tracing, and Critic-Gated Explanations

Tuan Dao Minh^a, Khanh-Trinh Nguyen^a, Duong Nguyen Tien^a, Quoc Khanh Ngo^b, Van-Hau Nguyen^a, Le Hoang Son^{b,*}

^a*Department of Quality Assurance and Testing, Hung Yen University of Technology and Education, Hung Yen, Vietnam*

^b*Artificial Intelligence Research Center, VNU Information Technology Institute, Vietnam National University, Hanoi, Vietnam*

Abstract

Graph-augmented knowledge tracing (KT) often skips leakage audits of the auxiliary knowledge it constructs, then reports a single AUC as if every graph component contributed. We present DH²A-KT, a two-tier knowledge system that is tractable on a single consumer GPU. **Tier 1 (DH²-KT)** reuses the leakage-controlled, train-only protocol of companion paper P0, extends it with hyperedge-level diagnostics, and then *ablates which knowledge predicts*. Registered gates ($\Delta\text{val} \geq +0.002$ vs. a matched twin) show that bipartite question–KC incidence and inter-step timing carry next-step accuracy, whereas hypergraph convolution, hint-item hyperedges, a full Q-matrix, teacher grouping, an expert DAG, and per-concept exponential forgetting do not. On XES3G5M fold 0 under a clean protocol (mask-repeats, chunked windows, $L=400$, test-only), the recommended tracer — a v4 LSTM with $Q \leftarrow KC$ incidence and a linear $\log(1+\Delta t)$ branch — reaches test AUC 0.8296

*Corresponding author.

Email address: sonlh@vnu.edu.vn (Le Hoang Son)

versus matched-budget GIKT 0.8258 (learner-level bootstrap 95% CI of the difference excludes 0; $Q \leftarrow KC$ alone is 0.8270). The same recipe on folds 1–2 stays within 0.001 (mean 0.8295 ± 0.0004). Timing is attempt context (query at $t+1$), not per-concept forgetting. **Tier 2 (AgentDG-KT)** is a Critic-gated agent layer over *frozen* Tier 1 scores. On the recommended tracer, grounded vs. independent-generation flag rates are 0.018/0.766 with mean ID-anchored KC-Jaccard 0.161/0.152 ($n=500$). Dual-rater Likert faithfulness 4.01 ± 1.07 / usefulness 3.52 ± 0.71 ($n=40$) was collected on the graph-only v2 checkpoint, not on that tracer. A limited observational hint ATE on FoundationalASSIST is reported as an identification check with weak propensity overlap, not as a primary claim. Graph-only v2 AUC 0.752 and three-fold manipulation checks are a forced-reliance diagnostic in the appendix, not the recommended tracer.

Keywords: Knowledge tracing, graph-structured knowledge, data leakage, large language model agents, explanation faithfulness, educational data mining

1. Introduction

Knowledge tracing (KT) estimates a learner’s latent mastery of knowledge components (KCs) from interaction history, supporting adaptive practice, intelligent tutoring, and curriculum recommendation. The field has progressed from probabilistic and sequence models [1] to graph-augmented models that propagate information over auxiliary concept graphs [2, 3], and, most recently, toward hypergraph representations that capture higher-order relations among knowledge components [4, 5, 6] and dynamic, temporally-

evolving graph structure [7].

This paper builds on [8]’s train-only leakage protocol. Tier 1 constructs and audits graph-structured knowledge, then asks which artefacts change next-step AUC under a registered gate. The recommended tracer is a GIKT-family v4 LSTM with $Q \leftarrow KC$ incidence and a linear Δt branch (Section 4.3); hypergraph convolution fails that gate (Section 6.3). Teacher, forum, and video members remain interfaces. Tier 2 explains frozen scores with a Critic (Section 5). Fig. 1 is the pipeline.

1.1. Motivation

Three concrete gaps motivate this work:

- **Higher-order structure is under-audited in graph-KT.** Hypergraph-KT models [4, 5, 6] group multiple knowledge components into a single hyperedge, yet published systems rarely audit whether those hyperedges were constructed without leakage, and few expose an interface for entities beyond KCs (hints, teachers, forums, video) when such logs become available.
- **Graph provenance is rarely audited.** [8] shows that pooling a knowledge-component graph from the full interaction log before a learner-based train/test split lets held-out information reach a graph-augmented KT model even when the sequence optimizer itself trains only on the training fold. [8]’s protocol audits this channel for *pairwise* prerequisite/similarity edges; no published audit protocol currently covers *hyperedges*.

- **Explanations of KT scores are rarely gated against a frozen predictor.** LLM rationales can ignore the numeric score they purport to explain. A knowledge system needs a leakage-closed explanation layer over frozen $P(\text{correct})$, not a second model that re-infers mastery.

1.2. Research Questions

This work is motivated by a broader question: *which leakage-audited knowledge actually improves or explains next-step KT?* We decompose this into three questions, plus entities this paper does not claim to have learned:

- **RQ1** (§6.3): Under a leakage audit, which structural knowledge improves next-step AUC?
- **RQ2** (§7.2): Does temporal knowledge contribute as forgetting of latent mastery, or as attempt-context at the scored step?
- **RQ3** (§7.5): Can a Critic-gated LLM-agent layer explain frozen Tier 1 scores without reopening leakage?

An observational hint ATE on FoundationalASSIST is reported as a secondary identification check (Section 7.6), not as a fourth research question. Interactions involving *teacher*, *forum/discussion*, and *video* nodes are defined in the schema (Section 3) but remain data-blocked on every corpus we use (Section 8.3): this paper establishes an interface for those entities, not empirical evidence of how they improve KT. A graph-only encoder and three-fold manipulation check remain available as a forced-reliance diagnostic (Appendix Appendix A); they are not RQ1’s recommended system.

1.3. Contributions

- **C1 (construction and audit).** A procedure that groups [8]’s audited, acyclic prerequisite edges into concept-prerequisite chain hyperedges and names three leakage classes a pairwise audit cannot detect (Section 4.2); session+Hint hyperedges on FoundationalASSIST [9] (Section 7.6). Discussion-thread and teacher-intervention types remain data-blocked (Section 8.3). Construction is not claimed as an AUC lift.
- **C2 (which knowledge predicts).** A pre-registered ablation (gate $\Delta\text{val} \geq +0.002$ vs. a matched twin) plus a clean-protocol tracer: $Q \leftarrow KC$ incidence passes ($\Delta\text{val}+0.00612$); a Linear $\log(1+\Delta t)$ branch on that backbone passes ($\Delta\text{val}+0.00280$; test AUC 0.8296 on XES3G5M fold 0, mean 0.8295 ± 0.0004 across three folds). HypergraphConv, hint-item hyperedges, a full Q-matrix, teacher grouping, an expert DAG, and per-concept forgetting fail (Table 6). Versus matched-budget GIKT 0.8258, the recommended tracer’s learner-level bootstrap 95% CI excludes 0 (Table 4). Graph-only v2 mean AUC 0.752 versus GKT 0.834 is an appendix diagnostic, not the recommended tracer.
- **C3 (Critic-gated explanations).** A Tier 2 pilot (AgentDG-KT, Section 5) over frozen Tier 1 scores at $n=500$ on a consumer GPU. On the recommended tracer, grounded vs. independent-generation flag rates are 0.018/0.766 (mean JC 0.161/0.152). Dual-rater Likert ($n=40$) is reported on the v2 checkpoint (Table 10). Comparison with [10] is a probability gate, not synonym-aware KC overlap or an AUC contest

— pilot evidence, not a classroom RCT.

2. Related Work

2.1. Graph- and hypergraph-based knowledge tracing

Pairwise KT graphs [2, 3] and dynamic variants [7] propagate over edges; hypergraph KT [4, 5, 6] and heterogeneous node types [11, 12] still draw members from KC/exercise/learner sets. None treat Teacher, Hint, Forum, or Video as first-class hyperedge members with KCs — C1’s *interface*. This paper’s quantitative core is a GIKT-class LSTM with audited $Q \leftarrow KC$ incidence (Section 4.3); hypergraph operators are registered failures (Table 6).

2.2. LLM agents and explanation faithfulness

Closest in scope is [10], which aligns LLM mastery reports with implicit DLKT states and evaluates explanation faithfulness with KC-overlap, an independent-generation control, and paired ablations. AgentDG-KT (Section 5) complements that line: we audit the auxiliary graph before explanation, gate rationales with a Critic over *frozen* $P(\text{correct})$, and use ID-anchored KC-Jaccard because XES3G5M has no in-repo synonym KC vocabulary. Broader LLM-agent surveys [13, 14, 15, 16] motivate the frozen-core split: re-inferring mastery with a multi-agent LLM over millions of interactions is not tractable on one consumer GPU.

2.3. Leakage-controlled evaluation

[8] (companion P0, under review) shows that pooling a KC graph from the full log before a learner split leaks held-out information through the graph even if the sequence optimizer trains only on the training fold. We reuse

its train-only construction, four pairwise diagnostics, and pinned artefacts (Section 3.1). This paper extends the audit from pairwise edges to hyperedges (Section 4.2).

Observational treatment-effect estimators in education [17, 18, 19] are background only: our IPW hint ATE is a secondary identification check, not a primary contribution (Section 4.4).

3. Preliminaries

Let $\mathcal{D} = \{(u, t, q, c, y)\}$ be a KT interaction log with learner u , timestamp t , item q , knowledge component c , and correctness label $y \in \{0, 1\}$, following [8]’s notation. A learner-based temporal split $\mathcal{F} = (\mathcal{F}_{\text{train}}, \mathcal{F}_{\text{val}}, \mathcal{F}_{\text{test}})$ assigns each learner to exactly one fold (ratios 0.7/0.1/0.2, three folds with split seeds 42/43/44, matching [8]’s protocol so results remain directly comparable, see Section 6.1).

3.1. Self-contained summary of the reused P0 protocol

Because [8] is under review, we state the reused pieces here. For each learner-based fold, E_{pre}^f and E_{sim}^f are built from *train-fold interactions only*, then cycle-pruned to a DAG. Four pairwise diagnostics (P0 Table of leakage metrics; computed on the train-only graph vs. held-out learners) are:

- ECR^{flag} : structural learner-overlap indicator (whether any learner appears in more than one split);
- $\text{ECR}^{\text{overlap}}$: held-out pattern overlap on retained edges;
- $|\rho|$: magnitude of the Pearson correlation between edge weight and held-out outcome;

- TBMR (P0 also writes TBVR): *within-train* temporal mixing — the share of post-cutoff *training* mass that supports retained prerequisite transitions — *not* a train/test membership violation.

P0 reports that switching from full-log to train-only graph construction shifts baseline AUC by at most 0.003 on XES3G5M, ASSISTments 2012, and Junyi. We pin the companion repository to commit `5e3d6a0` and copy its published baseline CSVs verbatim (Section 6.2). The companion PDF can be attached as journal supplementary material on request. Chain hyperedges in this paper are bounded-length paths on the same audited E_{pre}^f (Section 4.1).

We extend this notation with a hyperedge $h = (\kappa, \{(\tau_i, e_i)\}, f)$: kind $\kappa \in \{\text{concept-prerequisite, session, discussion-thread, teacher-intervention}\}$, typed members (student, exercise, concept, teacher, hint, forum, discussion, video), and fold f . Chain hyperedges on audited E_{pre} are audited and used in FAIL/diagnostic arms; they do *not* enter the recommended tracer (Table 4). Teacher/forum/video kinds are interfaces; session write-backs appear only in the Tier 2 pilot. The recommended tracer consumes train-only Q←KC incidence and timestamps inside a v4 LSTM (Section 4.3).

4. Tier 1: DH²-KT

4.1. Heterogeneous hypergraph construction

Concept-prerequisite hyperedges are bounded-length simple directed paths on audited, acyclic E_{pre}^f (Algorithm 1).

- 1: **Input:** E_{pre}^f ; bounds $[\ell_{\min}, \ell_{\max}]$
- 2: **Output:** chain hyperedge set $\mathcal{E}_f^{\text{cprereq}}$
- 3: Build directed graph G from E_{pre}^f

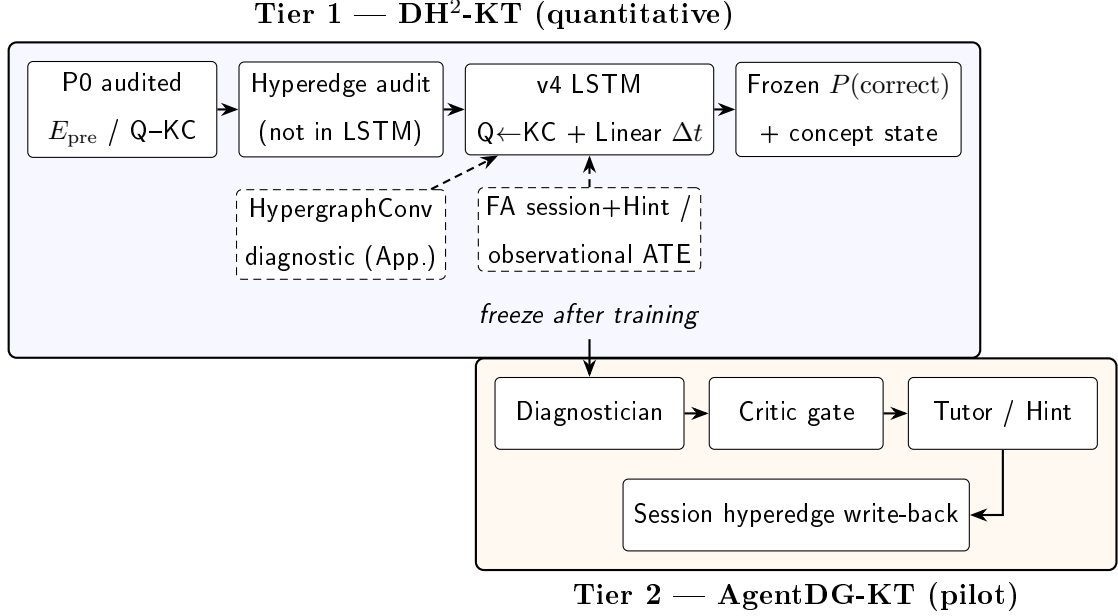


Figure 1: Two-tier DH²A-KT architecture. Solid Tier 1 path: leakage-audited construction, then the recommended v4 LSTM with $Q \leftarrow KC$ incidence and Linear Δt (no concept-stack HypergraphConv). Dashed: graph-only HypergraphConv diagnostic (Appendix Appendix A) and FoundationalASSIST session+Hint / observational ATE (secondary). After freezing, Tier 2 agents explain and gate those outputs: Diagnostician $\rightarrow$ Critic $\rightarrow$ Tutor, with session hyperedge write-back at pilot scale.

```

4: for each root  $r$  with in-degree 0 in  $G$  do
5:   for each simple path  $p$  from  $r$  with  $\ell_{\min} \leq |p| \leq \ell_{\max}$  do
6:     emit members  $\{(\text{concept}, c) : c \in p\}$ 
7:   end for
8: end for

```

Algorithm 1. Chain hyperedges from audited E_{pre}^f .

Discussion-thread and teacher-intervention hyperedges are data-blocked on the public KT corpora we use. Session+Hint hyperedges are built on

FoundationalASSIST only (Section 7.6). Chain hyperedges on XES3G5M are audited and used in the graph-only diagnostic (Appendix Appendix A), not as the recommended tracer’s message-passing graph.

4.2. Hyperedge-level leakage audit

The four pairwise diagnostics of Section 3.1 are reused on the pairwise projection of a hyperedge set. Pairwise TBMR/ $|\rho|$ are skipped when that projection exceeds the implementation cap (Table 2). Table 1 adds three classes a pairwise audit cannot express.

Measured values on XES3G5M fold 0 appear in Table 2.

4.3. Recommended tracer: v4 LSTM with $Q \leftarrow KC$ and time

The recommended DH²-KT tracer is architecture v4 with **-question-graph** and **-no-graph** on the concept stack: concept-prerequisite hyperedges are *not* consumed by **HypergraphConv** on this path. Let $A \in \mathbb{R}^{N_q \times N_c}$ be the train-only question–KC incidence, row-normalized to $\bar{A}$. Each item q has an embedding e_q ; each KC c has e_c . One propagation step, in the spirit of GIKT [3] rather than hypergraph convolution, is

$$\tilde{e}_q = \tanh(W(e_q + \bar{A}_{q,:} E_c)), \quad (1)$$

with a dedicated linear W so the mix cannot collapse into e_c (registered twin: same LSTM without this branch). Interaction embeddings of (c_t, y_t) plus Rasch-style item parameters feed an LSTM. The next step is scored by reading a query vector at the *known* identity of item/KC $t+1$ (standard next-step KT: the model is told *what* will be asked, not whether the learner will be correct).

When `-time-gap` is on (mode `both` on XES3G5M), a dedicated linear map of $\log(1 + \Delta t)$ is added on the LSTM input and on that query, where Δt is the inter-step gap aligned with the row being encoded. The query-side gap is the wait until the scored item; that item’s identity is part of the prediction request, so the gap is attempt context rather than a post-hoc peek at duration after y_{t+1} is observed. We do not feed the scored step’s own response into the query. Per-concept exponential forgetting is a registered alternative that fails the gate (P4). HypergraphConv on multi-KC item hyperedges, hint-item hypergraphs, dual-gated graph-only x_t , and the graph-sensitivity auxiliary loss L_{aux} belong to the v2 diagnostic encoder (Appendix Appendix A) or to FAIL arms (Table 6), not to Eq. (1).

4.4. *Observational ATE (secondary)*

On FoundationalASSIST we report a standard inverse-propensity (IPW) average treatment effect of `hint_usedt` on `correctt+1`, with logistic propensity on pre- t summaries and clipping documented in Section 7.6. Unconfoundedness is assumed, not proven. Same-step `discrete_score` is not the outcome (it is mechanically tied to hint requests). No ATE is reported on XES3G5M (no hint logs). This check is not a primary contribution.

Table 1: Hyperedge-level leakage taxonomy (extends [8] Table 1)

Class	Source	Diagnostic
Group-membership	A hyperedge with ≥ 1 member drawn from a held-out interaction while other members are train-only	Group-membership leak rate (XES3G5M fold 0: 0; Table 2)
Cross-modal	A Forum/Video embedding pooled over content timestamps after the prediction cutoff	Not instantiated (no Forum/Video logs on the corpora we train)
Intervention-timing	A Teacher/Hint hyperedge whose content was effectively chosen using knowledge of the student's later correctness	Not instantiated for Teacher; Hint members on FoundationalASSIST use train-only session construction (group-membership leak 0)

Table 2: Concept-prerequisite hyperedge audit on XES3G5M fold 0. Pairwise TBMR/ $|\rho|$ are skipped when the pairwise projection exceeds the implementation cap of 100,000 pairs (11,568,053 projected pairs in this fold; see audit notes).

Metric	Value
Hyperedges	446,305
ECR ^{flag}	0.0
Group-membership leak rate	0.0

5. Tier 2: AgentDG-KT pilot

Tier 2 is deliberately scoped as a pilot, not a benchmark at Tier 1’s quantitative rigor: a full multi-agent LLM pipeline that re-infers knowledge state over millions of interactions is not tractable on a single consumer GPU within a realistic project timeline. Three agents operate on Tier 1’s *frozen* output rather than re-inferring knowledge state:

- **Diagnostician Agent** produces a natural-language rationale for a given Tier-1 prediction, explicitly instructed not to override the given probability.
- **Tutor/Hint Agent** proposes the next hint/exercise and writes a new session hyperedge back into the graph (the agentic closed loop).
- **Critic Agent** is a *mandatory* quality gate that flags any Diagnostician output contradicting or ignoring the Tier-1 number it was meant to explain, before anything is treated as ground truth in a case study.

We ran a 500-sample pilot on XES3G5M fold 0 (seed 42, Qwen2.5-7B via Ollama) on the graph-only v2 checkpoint and re-ran the same prompt on the recommended tracer (Section 4.3). The Diagnostician emits a **GroundedKCs** trailer for ID-anchored KC-Jaccard; IG omits frozen $P(\text{correct})$ from that prompt while the Critic still sees the true score. Flag rates and JC are in Section 7.5. The Tutor wrote 500 session hyperedges at pilot scale.

Table 3: Tier 2 Critic flag rates (XES3G5M fold 0, $n=500$, seed 42, Qwen2.5-7B via Ollama unless noted). Flag rate = fraction of Diagnostician outputs flagged by the mandatory Critic. v1–v2 precede the structured **GroundedKCs** trailer; the grounded/IG pair is the KBS faithfulness re-run (Table 9).

Run	Backend	Flag rate	Mean $P(\text{correct})$
Stub (sanity)	StubLLM	0.000 (0/500)	0.742
Ollama v1	qwen2.5:7b	0.124 (62/500)	0.746
Ollama v2 (format calib.)	qwen2.5:7b	0.000 (0/500)	0.753
Grounded + GroundedKCs	qwen2.5:7b	0.026 (13/500)	—
IG (w/o P in Diag.)	qwen2.5:7b	0.768 (384/500)	—

6. Experimental setup

6.1. Datasets

Following [8], XES3G5M [20] (18,066 learners, 7,652 items, 865 KCs, 6,413,353 interactions) is the primary predictive benchmark. ASSISTments 2012 GKT AUC from P0 is not reused as a comparison column. FoundationalASSIST [9] is a secondary fold with synchronized hints (Section 7.6), not interchangeable with classic ASSISTments 2012. Junyi supplies an expert DAG for structural overlap only (Section 7.4); we do not train the full tracer there.

6.2. Baselines reused from P0

We reuse [8]’s published results for BKT, DKT, AKT, simpleKT, GKT, GIKT, SKT, DyGKT, and DGEKT under train-only graph construction (vendored CSVs at commit [5e3d6a0](#)), except GIKT and AKT on XES3G5M, which we retrain under the clean $L=400$ protocol (Table 4). A config-

checksum script confirms the XES3G5M P0 config match. The companion PDF is available to reviewers as supplementary material on request.

6.3. Manipulation check

A hypergraph ablation is interpretable only if destroying the constructed hyperedges moves AUC (near-total node-drop $p=0.90$). The graph-only v2 encoder passes on all three XES3G5M folds (Appendix Appendix A); an earlier pairwise+exercise path was graph-inert ($\Delta\text{AUC}\approx 0.0003$). The recommended v4 tracer uses `-no-graph` and is not claimed to pass this destruction check.

7. Results

7.1. Which structural knowledge predicts (RQ1)

Table 4 reports *test-only* AUC on XES3G5M fold 0 under the clean protocol (mask-repeats, chunked windows, $L=400$). DH²-KT with $Q\leftarrow KC$ incidence reaches 0.8270 versus matched-budget GIKT 0.8258 and AKT 0.8221; the no-graph twin is 0.8202. The $Q\leftarrow KC$ vs. twin gate is $\Delta\text{val}+0.00612$ (PASS). Learner-level bootstrap (1000 resamples, 3614 learners) for $Q\leftarrow KC$ minus GIKT is $+0.00117$, 95% CI $[+0.00078, +0.00153]$ (excludes 0). Adding Linear $\log(1+\Delta t)$ reaches test AUC 0.8296 (minus GIKT: $+0.00378$, 95% CI $[+0.00342, +0.00413]$). We do not credit hypergraph convolution for these numbers: that arm is `-no-graph` on the concept stack. The same $Q\leftarrow KC+\Delta t$ recipe on folds 1 and 2 stays within 0.001 of fold 0 (mean 0.8295 ± 0.0004 ; Table 5).

Table 6 lists registered failures (Δval gate $+0.002$, not loosened). HypergraphConv on multi-KC item hyperedges (K), hint-item session hyperedges

Table 4: Test-only AUC on XES3G5M fold 0 under the *clean* protocol: mask-repeats, chunked windows, max seq. len. 400. DH²-KT uses a v4 LSTM with question embeddings; Q←KC is bipartite incidence (**-question-graph**), not hypergraph convolution; + Δt is a Linear $\log(1+\text{gap})$ branch into LSTM and next-step query (**-time-gap**, both). GIKT is matched to the DH² budget (batch 16, cap 30 epochs, patience 5; early-stop epoch 8). AKT keeps P0 hyperparameters at $L=400$. Learner-level bootstrap: Q←KC+ Δt minus AKT is +0.00745, 95% CI [+0.00703, +0.00792]. This is *not* the three-fold graph-only v2 table (Appendix Appendix A).

Model	Test AUC	n scored
DH ² -KT Q←KC + Δt	0.8296	1,093,755
DH ² -KT Q←KC	0.8270	1,093,755
GIKT (matched budget)	0.8258	1,093,720
AKT ($L=400$)	0.8221	1,093,720
DH ² -KT Q←KC off (twin)	0.8202	1,093,755

Table 5: Q←KC+ Δt test-only AUC on XES3G5M under the clean protocol (mask-repeats, chunked, $L=400$), one model per learner fold. Folds are P0’s three split seeds. Mean±SD is over folds, not over bootstrap resamples of fold 0.

Fold	Test AUC	n scored
0	0.8296	1,093,755
1	0.8298	1,092,664
2	0.8291	1,090,374
Mean±SD	0.8295±0.0004	—

on FoundationalASSIST (L), a full Q-matrix (M1), teacher grouping (M4), a Junyi expert DAG on a 5k-user subsample (M5), and per-concept exponen-

Table 6: Negative results under a pre-registered validation gate ($\Delta\text{val} \geq +0.002$ vs. a matched twin). All runs keep a separate Linear/embedding branch (no absorption into `concept_embed`). FAIL means the knowledge type is *not* credited as predictive. Hypergraph convolution and hint-item hyperedges are among the failures; they are not AUC novelty.

ID	Knowledge (corpus)	Twin val	ON val	Δval	Gate
K	HypergraphConv, multi-KC item HE (XES)	0.82598	0.82623	+0.00025	FAIL
L	Hint-item session HE (FA)	0.80861	0.80868	+0.00007	FAIL
M1	Full Q-matrix incidence (FA)	0.80834	0.80824	-0.00010	FAIL
M4	Teacher grouping (ASSIST2012)	0.76435	0.76577	+0.00142	FAIL
M5	Expert prerequisite DAG (Junyi 5k)	0.7640	0.76456	+0.00056	FAIL
P4	Per-concept exp. forget vs. Linear Δt (XES)	0.82878	0.82620	-0.00258	FAIL

tial forgetting (P4) all FAIL. The small FoundationalASSIST session-HE shift $0.720 \rightarrow 0.722$ (concept projection; Table 7) is therefore not a hypergraph-AUC claim: the isolated hint-item HE arm did not pass.

7.2. Time as attempt context (RQ2)

On XES3G5M, Linear Δt into both LSTM and next-step query passes against $Q \leftarrow KC$ ($\Delta\text{val} + 0.00280$; test-only 0.8296 vs. 0.8270, difference +0.00261, learner-level 95% CI [+0.00225, +0.00298]). A quartile slice of $\log(1 + \text{gap})$ collapses to two bins: short gaps ($\lesssim 18$ min) $\Delta + 0.0028$, long gaps $\Delta + 0.0019$, both CIs excluding 0 — not a long-horizon forgetting signature. On FoundationalASSIST, query-only Δt keeps most of the M2 gain (+0.01471 vs. off) while LSTM-only is +0.00280; combining time with prior-step `saw_answer` is additive (+0.00297 vs. M2). Per-concept $\exp(-\text{softplus}(\theta_c) \cdot \text{gap})$ fails versus Linear (P4). We treat time as attempt-context knowledge, not as a

hypergraph effect. Query-side Δt uses the gap to the item whose identity is already the next-step query; it is not the duration of a step whose correctness has been observed. Against AKT at the same $L=400$ clean protocol (test AUC 0.8221), $Q \leftarrow KC + \Delta t$ is ahead by +0.00745 (learner-level 95% CI [+0.00703, +0.00792]).

7.3. Paired diagnostic knobs

Table 7 pairs remaining diagnostic knobs with Tier 2 faithfulness knobs. Removing the Critic sets the flag rate to 0 by construction while leaving ID-anchored KC-Jaccard unchanged; IG mainly inflates the Critic flag rate ($0.026 \rightarrow 0.768$) with only a small JC shift. Dropping L_{aux} (`graph_sensitivity_weight = 0`) on the graph-only fold 0 track still yields clean AUC 0.754 and $\Delta\text{AUC} = 0.039$ (pass); that track is the appendix diagnostic, not Table 4.

7.4. Ground-truth structural overlap (Junyi)

Table 8 compares hyperedge-chain overlap with P0’s pairwise E_{pre} against the Junyi expert DAG at $K=|\text{expert}|$. Edge F1 is 0.165 (hyperedge chain) vs. 0.183 (E_{pre}) on fold 0; folds 1–2 are analogous. The chain representation slightly *underlaps* pairwise E_{pre} : collapsing directed paths into multi-node hyperedges, then projecting back to pairs for overlap scoring, can dilute expert-edge recall relative to scoring E_{pre} directly. We treat this as structural validation of construction fidelity, *not* as evidence that chain hyperedges improve predictive KT on Junyi — full DH²-KT training on Junyi (25M interactions) was not run.

Table 7: Paired ablation of DH²A-KT knowledge and explanation knobs. Predictive rows use the clean $L=400$ protocol (fold 0, test-only) except HypergraphConv / FA session (validation or secondary corpus, as marked). The registered gate is $\Delta\text{val} \geq +0.002$; FAIL rows are not AUC novelty. Manipulation ΔAUC is the v2 graph-only diagnostic (Appendix Appendix A), not the recommended tracer.

Setting	AUC	Gate / note	Flag	Mean JC
Q $\leftarrow$ KC + Δt (recommended)	0.8296	PASS $\Delta\text{val}+0.00280$	—	—
Q $\leftarrow$ KC on	0.8270	PASS $\Delta\text{val}+0.00612$	—	—
Q $\leftarrow$ KC off (twin)	0.8202	—	—	—
HypergraphConv (val)	0.8262	FAIL +0.00025	—	—
Grounded + Critic (v2 ckpt)	0.752 $\pm$ 0.002	manip. 0.038–0.047	0.026	0.152
IG (v2)	—	—	0.768	0.142
Grounded + Critic (v4 + Δt)	0.8296	same prompt, $n=500$	0.018	0.161
IG (v4 + Δt)	—	—	0.766	0.152
w/o Critic	—	—	0.000	0.152
FA: concept-only	0.720	—	—	—
FA: + session HE	0.722	L FAIL +0.00007	—	—

Table 8: Ground-truth cross-validation on Junyi Academy (fold 0, top- $K=|\text{expert}|=1131$). Hyperedge chain slightly underlaps pairwise E_{pre} but remains the same order of magnitude.

Representation	Edge F1	Edge prec.	Edge rec.
P0 E_{pre}	0.183	0.183	0.183
Hyperedge chain	0.165	0.165	0.165

7.5. Tier 2 faithfulness pilot (RQ3)

See Table 3 and Section 5. The Tutor wrote 500 session hyperedges at pilot scale. Following [10]’s use of a corpus-wide KC-overlap faithfulness met-

ric and an independent-generation negative control, we additionally compute *ID-anchored KC-Jaccard* over Diagnostician outputs: the model must emit a structured **GroundedKCs**: [...] trailer; support is the target concept, KCs in the recent history window, and their 1-hop neighbors on audited train-only E_{pre} . The IG arm uses the same history but omits frozen $P(\text{correct})$ from the Diagnostician prompt while the Critic still sees the true Tier 1 score (Table 9). On Qwen2.5-7B (Ollama, $n=500$), grounded explanations have flag rate 0.026 and mean JC 0.152; IG raises the Critic flag rate to 0.768 while mean JC stays near 0.142 ($\Delta\text{JC} \approx +0.010$), so the gate mainly catches probability-ungrounded rationales rather than rewriting KC overlap. The same prompt on the recommended $Q \leftarrow KC + \Delta t$ checkpoint (fold 0, $n=500$) reproduces the pattern: grounded/IG flag 0.018/0.766, mean JC 0.161/0.152. StubLLM harness rows ($n=20$) validate the pipeline (flag 0.0 vs. 1.0 at matched JC). From the calibrated Ollama v2 run we drew a fixed 40-sample pack (seed 42) for two independent raters (A1, B1) on Likert faithfulness and usefulness (1–5). Pooled means are 4.01 ± 1.07 (faithfulness) and 3.52 ± 0.71 (usefulness); pairwise Spearman $\rho=0.71$ / 0.58 and within- ± 1 rates of 92.5% / 97.5% (Table 10). A third rater was optional and not required for this pilot report.

7.6. Secondary corpus: FoundationalASSIST

FoundationalASSIST [9] provides $\approx 1.74\text{M}$ skill-linked interactions with synchronized `hint_count` (hint-use rate 5.2%). Using the same learner-temporal protocol as P0 (0.7/0.1/0.2, fold 0), we infer train-only E_{pre} (280 edges after cycle pruning), build 5,840 concept-prerequisite chain hyperedges, and construct 308,670 session hyperedges on train users (11.6% with a Hint

Table 9: Tier 2 explanation faithfulness: grounded Diagnostician vs. independent generation (IG). ID-anchored KC-Jaccard (JC) measures overlap between **GroundedKCs** trailer IDs and the support set $\{\text{target}\} \cup \text{history} \cup \text{1-hop } E_{\text{pre}}$. IG omits frozen $P(\text{correct})$ from the Diagnostician prompt; the Critic still sees the true Tier-1 score. The original Ollama pack used the v2 graph-only checkpoint (also the human-rating pack). The recommended $Q \leftarrow KC + \Delta t$ tracer is a same-prompt re-run ($n=500$).

Arm	Backbone	Backend	n	Flag	Mean JC
Grounded	—	StubLLM	20	0.000	0.191
IG	—	StubLLM	20	1.000	0.191
Grounded	v2 graph-only	qwen2.5:7b	500	0.026	0.152
IG	v2 graph-only	qwen2.5:7b	500	0.768	0.142
Grounded	v4 $Q \leftarrow KC + \Delta t$	qwen2.5:7b	500	0.018	0.161
IG	v4 $Q \leftarrow KC + \Delta t$	qwen2.5:7b	500	0.766	0.152

Table 10: Human rating of Tier 2 outputs (XES3G5M fold 0 Ollama v2 pilot subsample, $n=40$ explanations, 2 independent raters, Likert 1–5). Agreement is pairwise Spearman ρ and fraction of scores within ± 1 .

Scale	Mean $\pm$ SD	Agreement (A1 vs. B1)
Faithfulness	4.01 ± 1.07	$\rho=0.71$; within ± 1 : 92.5%
Usefulness	3.52 ± 0.71	$\rho=0.58$; within ± 1 : 97.5%

member; $\text{ECR}^{\text{flag}}=0$, group-membership leak=0). DH²-KT fold 0 AUC is 0.720 with concept-prerequisite hyperedges alone ($n=298,500$ predictions). Adding up to 20k multi-concept session hyperedges (Hint-preferring selection; concept co-occurrence projection) raises AUC to 0.722 ($\Delta+0.002$) — a small positive shift under the same local budget, not a matched GKT comparison. Using the causal layer on train transitions, the IPW ATE of

hint use on *next-step* correctness is -0.207 (naive mean difference -0.255 ; trimmed IPW -0.211 ; bootstrap 95% CI $[-0.230, -0.187]$ on 50k-row re-samples; $n_{\text{treated}}=36,151$). Propensity scores hit the clip floor ($[0.05, 0.219]$), indicating weak overlap: struggling learners disproportionately request hints, so the negative association is *not* interpretable as a proven causal penalty of giving hints. Table 11 summarizes.

Table 11: FoundationalASSIST secondary corpus (fold 0). Session hyperedges use a 30-minute gap; Hint members appear when `hint_count` > 0. Predictive AUC is observational (no P0-reused GKT/simpleKT column). Hint ATE is IPW of `hint_usedt` on `correctt+1` under pre- t confounders; propensity mass hits the clip floor — treat as limited observational evidence.

Quantity	Value
Interactions (canonical)	1,740,987
Learners / items / KCs	5,000 / 3,359 / 209
Hint-use rate	0.052
Train-only E_{pre} edges (after cycle prune)	280
Concept-prerequisite chain hyperedges	5,840
Session hyperedges (train users)	308,670
with ≥ 1 Hint member	35,752 (11.6%)
Session hyperedge ECR^{flag} / group leak	0 / 0
DH ² -KT fold 0 AUC (concept-prereq)	0.720
+ session HE (20k, Hint-prefer)	0.722 ($\Delta + 0.002$)
n next-step predictions	298,500
Hint ATE (IPW) / naive mean diff	−0.207 / −0.255
trimmed IPW (propensity 10–90%)	−0.211
bootstrap 95% CI (subsample)	[−0.230, −0.187]
n treated / control (train transitions)	36,151 / 660,321
propensity [min, max]	[0.05, 0.219]

8. Discussion

8.1. Threats to validity

Headline AUCs (Table 4) are clean $L=400$, test-only. $Q \leftarrow KC + \Delta t$ is also on folds 1–2; matched-budget GIKT/AKT remain fold 0. Other baselines come from reused P0 tables on the older budget. The GIKT edge is small; we report learner-level bootstrap CIs, not a SOTA headline. HypergraphConv and hint-item hyperedges fail the registered gate, so we do not credit multi-way message passing. The v2 destruction check (Appendix Appendix A) shows that *a* backbone can be forced to use a hypergraph; it does not identify which row of Table 6 drives the recommended tracer.

8.2. What KBS should take from a small AUC edge

DH²-KT is a leakage-audited knowledge system, not “hypergraph NNs beat GIKT”: (i) hyperedge diagnostics beyond pairwise E_{pre} ; (ii) a registered map of which knowledge predicts; (iii) a Critic gate over frozen $P(\text{correct})$, vs. [10] as a probability gate rather than synonym-aware KC overlap. The ≈ 8 point gap versus GKT is the cost of the graph-only diagnostic, not of Table 4. The hint ATE is a weak-overlap check, not a fourth pillar. Dual-rater Likert ($n=40$) is on v2; automatic flag/JC in the abstract use the recommended tracer.

8.3. Limitations

- Discussion-thread and teacher-intervention hyperedges remain data-blocked; session+Hint hyperedges on FoundationalASSIST shift AUC $0.720 \rightarrow 0.722$ under concept projection, but the registered hint-item

hypergraph arm fails the +0.002 gate (L in Table 6) and is not claimed as hypergraph novelty.

- The causal-effect layer’s identification assumption (no unmeasured confounding given observed features) is stated, not proven; the FoundationalASSIST hint ATE shows weak propensity overlap and should not be read as a recommendation against hints.
- The hyperedge leakage audit’s pairwise-projection approximation (Section 4.2) weights all projected pairs equally; a hyperedge-native statistic is future work.
- Tier 2 pilot: one fold, one local 7B LLM; human ratings use $n=40$ and two raters on the $v2$ graph-only checkpoint (no third rater; not a classroom deployment study; not the recommended tracer). Automatic flag/JC numbers in the abstract use the $Q \leftarrow KC + \Delta t$ re-run. ID-anchored KC-Jaccard is not synonym-aware (XES3G5M lacks an in-repo KC name vocabulary); the grounded-IG ΔJC is small ($\approx +0.010$), so KC-overlap alone is a weak separator relative to Critic flag rate. An LLM-scale sensitivity sweep remains optional future work.
- Reproduction environment: Windows workstation, single RTX 3090; unit tests pass in the released codebase.

9. Conclusion and future work

We presented DH²A-KT: leakage-audited construction, a registered map of which knowledge predicts ($Q \leftarrow KC$ and Linear Δt live; hypergraph convolution and several metadata sources fail), and a Critic-gated explanation

layer over frozen scores. On XES3G5M (clean $L=400$, test-only), the recommended tracer reaches 0.8296 on fold 0 versus matched-budget GIKT 0.8258 (0.8295 ± 0.0004 over three folds). Timing is attempt context, not concept-forget. Tier 2 grounded/IG flag rates on that tracer are 0.018/0.766. Future work: matched GIKT/AKT on folds 1–2, human ratings on the recommended tracer, hyperedge-native leakage statistics, and richer ATE confounders.

CRedit authorship contribution statement

Tuan Dao Minh: Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Khanh-Trinh Nguyen:** Validation, Data curation, Writing – review & editing. **Duong Nguyen Tien:** Validation, Data curation, Writing – review & editing. **Quoc Khanh Ngo:** Investigation, Resources, Writing – review & editing. **Van-Hau Nguyen:** Validation, Resources. **Le Hoang Son:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Code for DH²A-KT is available at <https://github.com/edu-risk-lab/DH2A-KT>. P0’s audited edges and reused baselines are vendored under

[8]’s release terms (pinned commit, Section 6.2); the companion PDF can be supplied as journal supplementary material while that manuscript is under review. FoundationalASSIST [9] must be obtained from Hugging Face under its Responsible Use Guidelines and cannot be redistributed by the authors.

Acknowledgment

All Tier 1 experiments in this paper are designed to run on a single consumer-grade NVIDIA RTX 3090 (24GB), matching the compute budget already validated by [8] on the same primary benchmarks. FoundationalASSIST [9] was accessed under the Hugging Face dataset’s Responsible Use Guidelines for this study.

Appendix A. Graph-only encoder as a forced-reliance diagnostic

The recommended tracer is v4 with $Q \leftarrow KC$ and Linear Δt (Table 4). A graph-only v2 encoder (chain hyperedges, no question LSTM path) exists so a destruction check is interpretable: mean test AUC 0.7516 ± 0.0015 vs. reused GKT 0.8336 ± 0.0010 under P0’s GKT budget ($L=200$; Table A.12). Frozen-weight node-drop $p=0.9$ passes on all three folds (ΔAUC 0.038/0.043/0.047); P0 GKT also drops after retraining on a destroyed pairwise graph (0.0877 ± 0.0009). Protocol differs (freeze vs. retrain). We do not deploy this encoder.

Table A.12: Diagnostic graph-only track (not the recommended system). Test AUC on XES3G5M under P0’s matched GKT budget (batch 4, 10 epochs, max seq. len. 200). DH²-KT v2 uses chain concept-prerequisite hyperedges and graph-only interaction. Mean±SD over three learner folds.

Fold	DH ² -KT v2	GKT (P0)	simpleKT (P0)
0	0.7501	0.8346	0.8744
1	0.7518	0.8338	0.8747
2	0.7531	0.8326	0.8750
Mean±SD	0.7516±0.0015	0.8336±0.0010	0.8747±0.0003
Δ vs. GKT (mean±SD)		−0.0820±0.0025	

Table A.13: Manipulation check at node-drop $p=0.9$ on XES3G5M. DH²-KT: train on clean hyperedges, re-evaluate frozen weights on destroyed hyperedges (folds 0–2). Training restores the best epoch by train loss (fold 1 previously collapsed on the final epoch). GKT: P0 DDR-downstream mean Δ AUC over $n=9$ fold–seed runs ([8]; retrain on destroyed pairwise graph). simpleKT has no concept-graph input, so graph destruction is undefined.

Model	AUC clean	Δ AUC ($p=0.9$)	DDR	Pass?
DH ² -KT v2 (fold 0)	0.753	0.0380	0.990	Yes
DH ² -KT v2 (fold 1)	0.749	0.0432	0.990	Yes
DH ² -KT v2 (fold 2)	0.753	0.0469	0.990	Yes
GKT (P0, mean±SD)	—	0.0877±0.0009	0.990	Yes
simpleKT (P0)	—	N/A	—	N/A

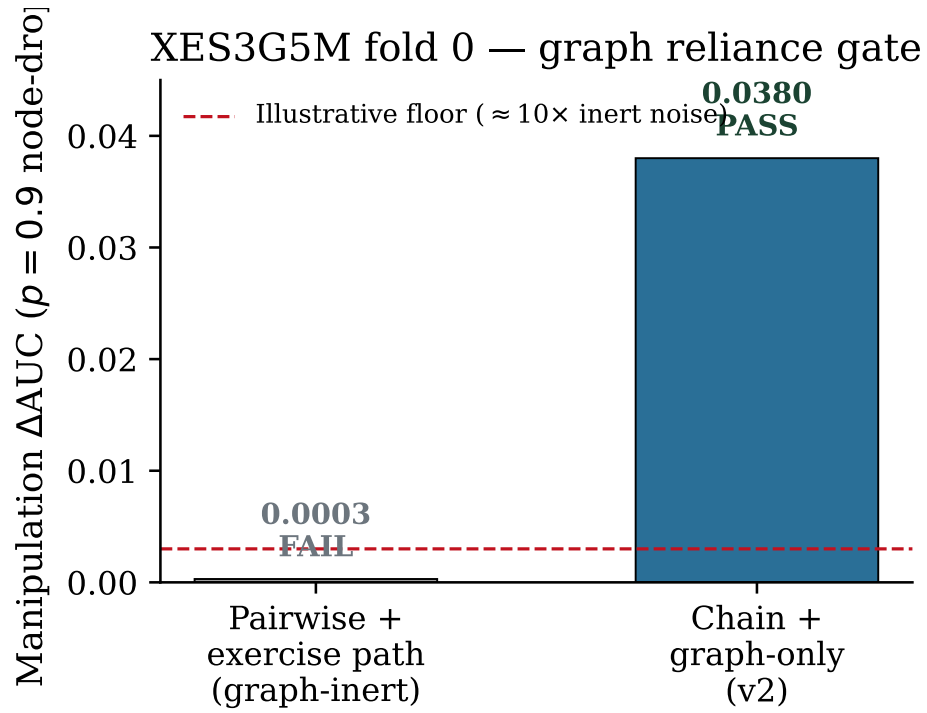


Figure A.2: Manipulation-check ΔAUC on XES3G5M fold 0 under $p=0.9$ node-drop. An early pairwise+exercise-embedding backbone is graph-inert ($\Delta\text{AUC} \approx 0.0003$); the chain+graph-only redesign passes ($\Delta\text{AUC} = 0.038$). The dashed line at 0.003 is an *illustrative* floor ($\approx 10\times$ the graph-inert noise), not a formal statistical cutoff from [8].

References

- [1] C. Piech, J. Bassen, J. Huang, S. Ganguli, M. Sahami, L. J. Guibas, J. Sohl-Dickstein, Deep knowledge tracing, in: Advances in Neural Information Processing Systems, Vol. 28, 2015, pp. 505–513. doi:10.48550/arXiv.1506.05908.
URL <https://doi.org/10.48550/arXiv.1506.05908>
- [2] H. Nakagawa, Y. Iwasawa, Y. Matsuo, Graph-based knowledge tracing: Modeling student proficiency using graph neural networks, in: IEEE/WIC/ACM International Conference on Web Intelligence (WI), 2019, pp. 156–163. doi:10.1145/3350546.3352513.
URL <https://doi.org/10.1145/3350546.3352513>
- [3] Y. Yang, J. Shen, Y. Qu, Y. Liu, K. Wang, Y. Zhu, W. Zhang, Y. Yu, GIKT: A graph-based interaction model for knowledge tracing, in: Machine Learning and Knowledge Discovery in Databases (ECML-PKDD), 2020, pp. 299–315. doi:10.1007/978-3-030-67658-2_18.
URL https://doi.org/10.1007/978-3-030-67658-2_18
- [4] M. Mohammadi, K. Berahmand, S. Sadiq, H. Khosravi, Knowledge tracing with a temporal hypergraph memory network, in: Artificial Intelligence in Education. Posters and Late Breaking Results, Workshops and Tutorials, Industry and Innovation Tracks, Practitioners, Doctoral Consortium, Blue Sky, and WideAIED, Vol. 2592 of Communications in Computer and Information Science, Springer, 2025, pp. 77–85. doi:10.1007/978-3-031-99267-4_10.
URL https://doi.org/10.1007/978-3-031-99267-4_10

- [5] F. Ma, C. Zhu, P. Lei, Hypergraph-driven high-order knowledge tracing with a dual-gated dynamic mechanism, *Applied Sciences* 15 (15) (2025) 8617. doi:10.3390/app15158617.
URL <https://doi.org/10.3390/app15158617>
- [6] X. Guo, X. Chang, X. Zhang, K. Liang, Y. Shang, Y. Zhang, HTKT: Knowledge tracing based on hypergraph transformer, in: *IEEE International Symposium on Parallel and Distributed Processing with Applications (ISPA)*, IEEE, 2024, pp. 1472–1477. doi:10.1109/ISPA63168.2024.00199.
URL <https://doi.org/10.1109/ISPA63168.2024.00199>
- [7] K. Cheng, L. Peng, P. Wang, J. Ye, L. Sun, B. Du, DyGKT: Dynamic graph learning for knowledge tracing, in: *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2024, pp. 409–420. doi:10.1145/3637528.3671773.
URL <https://doi.org/10.1145/3637528.3671773>
- [8] D. M. Tuan, K.-T. Nguyen, D. N. Tien, Q. K. Ngo, V.-H. Nguyen, L. H. Son, Leakage-controlled concept graph construction and cold-start diagnostic protocol for knowledge tracing, submitted to *Applied Intelligence (APIN)*; code and artefacts at <https://github.com/edu-risk-lab/leakage-controlled-kt-audit> (2026).
- [9] ASSISTments Foundation, FoundationalASSIST: An educational dataset for foundational knowledge tracing and pedagogical grounding of LLMs, arXiv:2602.00070, hugging Face dataset: <https://huggingface.co/datasets/ASSISTments/FoundationalASSIST> (2026).

doi:10.48550/arXiv.2602.00070.

URL <https://doi.org/10.48550/arXiv.2602.00070>

- [10] E. Ni, J. Wang, J. Luan, J. Liu, S. Xiong, S. Xu, W. Cheng, M. Chen, Y. Ni, Y. Li, Interpretable knowledge tracing via explicit–implicit alignment, *Knowledge-Based Systems* 344 (2026) 116071. doi:10.1016/j.knosys.2026.116071.
URL <https://doi.org/10.1016/j.knosys.2026.116071>
- [11] Z. Wang, W. Wu, C. Zeng, H. Luo, J. Sun, Psychological factors enhanced heterogeneous learning interactive graph knowledge tracing for understanding the learning process, *Frontiers in Psychology* 15 (2024) 1359199. doi:10.3389/fpsyg.2024.1359199.
URL <https://doi.org/10.3389/fpsyg.2024.1359199>
- [12] Q. Ni, T. Wei, J. Zhao, L. He, C. Zheng, HHSKT: A learner–question interactions based heterogeneous graph neural network model for knowledge tracing, *Expert Systems with Applications* 215 (2023) 119334. doi:10.1016/j.eswa.2022.119334.
URL <https://doi.org/10.1016/j.eswa.2022.119334>
- [13] Z. Chu, S. Wang, J. Xie, T. Zhu, Y. Yan, J. Ye, A. Zhong, X. Hu, J. Liang, P. S. Yu, Q. Wen, LLM agents for education: Advances and applications, in: *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025, pp. 13832–13870. doi:10.18653/v1/2025.findings-emnlp.743.
URL <https://doi.org/10.18653/v1/2025.findings-emnlp.743>

- [14] W. Zhou, H. Ma, X. Yu, C. Wang, S. Yang, X. Zhang, AgentCAT: Simulating computerized adaptive testing via multi-agent large language models, arXiv:2606.21832 (2026). doi:10.48550/arXiv.2606.21832.
URL <https://doi.org/10.48550/arXiv.2606.21832>
- [15] Y. Liu, G. Zhang, K. Wang, S. Li, S. Pan, Graph-augmented large language model agents: Current progress and future prospects, IEEE Intelligent Systems (2025). doi:10.1109/MIS.2025.3642667.
URL <https://doi.org/10.1109/MIS.2025.3642667>
- [16] X. Cheng, Z. Zhang, J. Wang, L. Fang, C. He, Q. Guan, S. Pan, W. Luo, GraphRAG-induced dual knowledge structure graphs for personalized learning path recommendation, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 40, 2026. doi:10.1609/aaai.v40i17.38479.
URL <https://doi.org/10.1609/aaai.v40i17.38479>
- [17] P. Kaur, A. Polyzou, G. Karypis, Causal inference in higher education: Building better curriculums, in: Proceedings of the Sixth (2019) ACM Conference on Learning @ Scale, 2019, pp. 1–4. doi:10.1145/3330430.3333663.
URL <https://doi.org/10.1145/3330430.3333663>
- [18] N. A. Kumar, W. Feng, J. Lee, H. McNichols, A. Ghosh, A. Lan, A conceptual model for end-to-end causal discovery in knowledge tracing, in: Proceedings of the 16th International Conference on Educational Data Mining (EDM 2023), 2023, pp. 413–418. doi:10.5281/zenodo.8

115727.

URL <https://doi.org/10.5281/zenodo.8115727>

- [19] S. Minn, J.-J. Vie, K. Takeuchi, H. Kashima, F. Zhu, Interpretable knowledge tracing: Simple and efficient student modeling with causal relations, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 36, 2022, pp. 12810–12818. doi:10.1609/aaai.v36i11.21560.

URL <https://doi.org/10.1609/aaai.v36i11.21560>

- [20] Z. Liu, Q. Liu, T. Guo, J. Chen, S. Huang, X. Zhao, J. Tang, W. Luo, J. Weng, XES3G5M: A knowledge tracing benchmark dataset with auxiliary information, in: Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track, 2023, pp. 32958–32970. doi:10.52202/075280-1429.

URL <https://doi.org/10.52202/075280-1429>